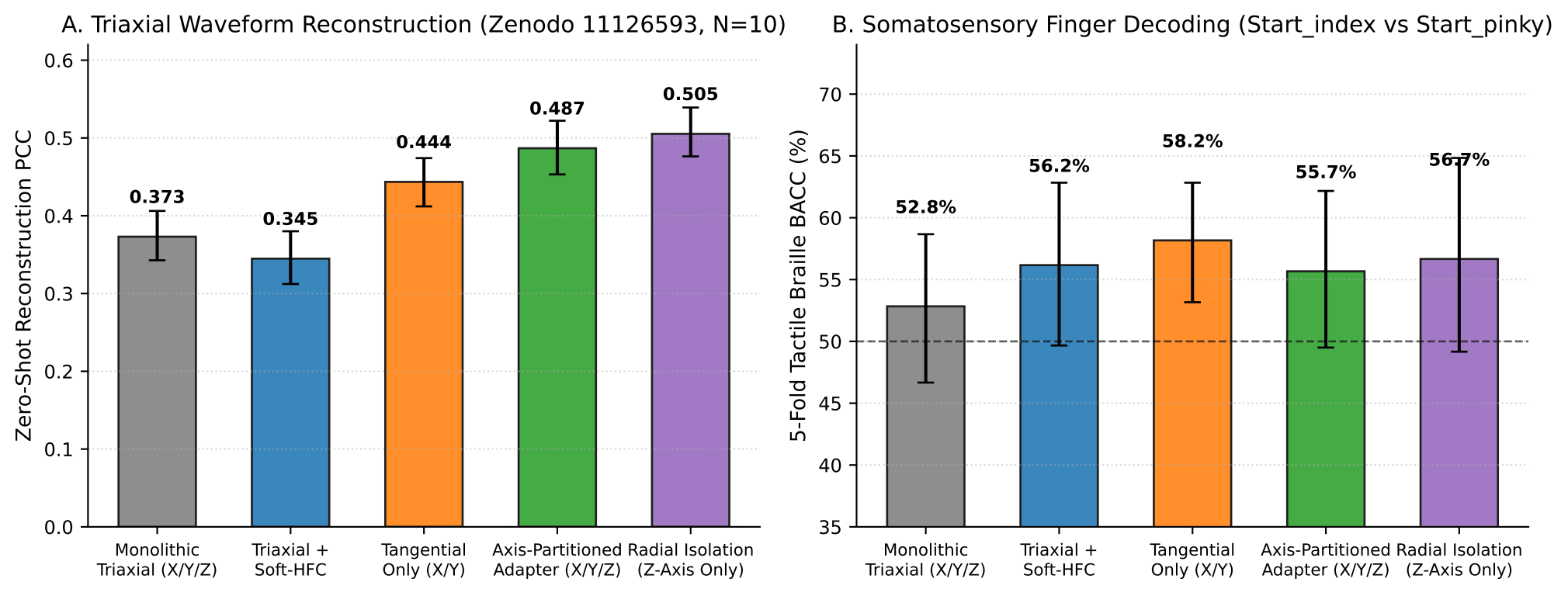
A. Triaxial Waveform Reconstruction (Zenodo 11126593, N=10)
Zero-Shot Reconstruction PCC
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.373
0.345
0.444
0.487
0.505
Monolithic Triaxial (X/Y/Z)
Triaxial + Soft-HFC
Tangential Only (X/Y)
Axis-Partitioned Adapter (X/Y/Z)
Radial Isolation (Z-Axis Only)
B. Somatosensory Finger Decoding (Start_index vs Start_pinky)
5-Fold Tactile Braille BACC (%)
35
40
45
50
55
60
65
70
52.8%
56.2%
58.2%
55.7%
56.7%
Monolithic Triaxial (X/Y/Z)
Triaxial + Soft-HFC
Tangential Only (X/Y)
Axis-Partitioned Adapter (X/Y/Z)
Radial Isolation (Z-Axis Only)